CHAPTER 1

REVIEW QUESTIONS

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1. What defines Big Data in modern digital environments?
   (a) Data that is always structured
   (b) Data that fits in traditional databases
   (c) Vast, fast, and complex data beyond traditional processing
   (d) Data stored only in cloud systems

2. Which 'V' of Big Data is concerned with the accuracy and reliability of data?
   (a) Volume
   (b) Veracity
   (c) Variability
   (d) Variety

3. Which of the following best represents the 3 Vs of Big Data?
   (a) Volume, Velocity, Variety
   (b) Volume, Validity, Variance
   (c) Variability, Volume, Verification
   (d) Velocity, Visualization, Veracity

4. In Big Data systems, MapReduce primarily improves performance by:
   (a) Increasing database normalization
   (b) Reducing transaction conflicts
   (c) Parallelizing computation across distributed nodes
   (d) Decreasing data redundancy

5. Which of the following Seaborn plots is used to visualize the relationship between two numerical variables?
   (a) Bar plot
   (b) Scatter plot
   (c) Box plot
   (d) Heatmap
6. Which Seaborn plot is best for showing the frequency of occurrences for each category in a categorical variable?
   (a) Count Plot
   (b) Line plot
   (c) Rug plot
   (d) Pair plot
7. What is the primary purpose of a violin plot in Seaborn?
   (a) To compare categorical data
   (b) To display the distribution and density of a dataset
   (c) To visualize relationships between two numerical variables
   (d) To create a heatmap of correlations
8. Which Seaborn plot helps visualize pairwise relationships between multiple numerical variables in a dataset?
   (a) Pair plot
   (b) Relational plot
   (c) Swarm plot
   (d) Cluster map
9. Which Seaborn function is best suited for visualizing the probability density of a continuous variable?
   (a) sns.boxplot()
   (b) sns.kdeplot()
   (c) sns.countplot()
   (d) sns.pointplot()
10. Which type of visualization is best for geographical data?

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(a) Bar Chart

(b) Line Chart

(c) Map or Heat Map

(d) Histogram

11. Which type of visualization is best for showing trends over time?

(a) Line chart

(b) Pie chart

(c) Heat map

(d) Box plot

12. What is a major challenge in data visualization?

(a) It always speeds up understanding

(b) Choosing the wrong chart can mislead viewers

(c) It eliminates the need for data cleaning

(d) It guarantees perfect insights

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13. What is the primary function of machine learning?

(a) Writing explicit rules for all tasks

(b) Learning from data to make predictions

(c) Storing large amounts of information

(d) Improving hardware efficiency

14. Which of the following is NOT a common application of machine learning?

(a) Image recognition

(b) Speech processing

(c) Cooking recipes

(d) Fraud detection

15. What is the key difference between supervised and unsupervised learning?

(a) Supervised learning requires labeled data, while unsupervised learning does not

(b) Supervised learning is always faster than unsupervised learning

(c) Supervised learning only works with numbers

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(d) Unsupervised learning gives exact answers while supervised learning does not

16. What is the role of reinforcement learning?

(a) Training machines to follow strict rules

(b) Learning through trial and error with rewards and penalties

(c) Only clustering data into groups

(d) Improving supervised learning models

17. How do machine learning models improve their performance over time?

(a) By manually changing the algorithm every time

(b) By continuously adjusting based on new data

(c) By deleting old data frequently

(d) By using a single dataset forever

18. What is one major challenge of machine learning?

(a) It requires no data to function

(b) It eliminates the need for human oversight

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(c) This can produce biased outcomes if trained on biased data

(d) It can only be used for gaming applications

19. Which of the following best describes supervised learning?

(a) Learning from labeled data to make predictions

(b) Finding hidden patterns in unlabeled data

(c) Analyzing data without any prior training

(d) Making decisions without any data

20. Which of the following is an example of unsupervised learning?

(a) Predicting house prices using historical sales data

(b) Categorizing spam emails based on predefined labels

(c) Grouping customers based on purchasing behavior

(d) Identifying fraudulent transactions with labeled examples

21. Which statement best describes how a machine "learns" in machine learning?

(a) By following fixed instructions in a program

(b) By identifying patterns in data and improving through experience

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(c) By randomly guessing outputs until it succeeds

(d) By storing all data without any processing

22. Document Database is a database type of which of the following

(a) NoSQL

(b) SQL

(c) JSON

(d) None of the above

23. In NoSQL, what does "horizontal scalability" mean?

(a) Adding more servers or nodes

(b) Adding more powerful CPUs

(c) Running databases on GPUs

(d) Resizing the database tables horizontally

24. Which of the following is a reason to choose NoSQL over SQL for big data applications?

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(a) Need for complex joins

(b) Better ACID support

(c) Support for flexible schemas and horizontal scaling

(d) Better GUI tools

25. Which challenge relates to the difficulty in choosing a NoSQL database due to different models and features?

(a) Lack of performance

(b) Limited GUI tools

(c) Lack of standardization

(d) ACID compliance

26. Which of the following is NOT a key benefit of performing Exploratory Data Analysis (EDA)?

(a) Identifying data errors and outliers

(b) Selecting important features for modeling

(c) Automatically building predictive models

(d) Understanding data distribution and patterns

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27. Which type of EDA focuses on analyzing the relationship between two variables?
    (a) Univariate Analysis
    (b) Bivariate Analysis
    (c) Multivariate Analysis
    (d) Time Series Analysis

28. When handling missing data during EDA, which method involves filling missing values with the mean or median?
    (a) Deletion
    (b) Imputation
    (c) Normalization
    (d) Transformation

29. Which method is commonly used to detect outliers in numerical data?
    (a) Interquartile Range (IQR)
    (b) Cross-tabulation
    (c) One-hot Encoding
    (d) Correlation Matrix

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30. You discover a highly skewed numerical column. Which transformation is often used to reduce skewness for EDA or modeling?
    (a) Squaring the values
    (b) Taking the logarithm of the values
    (c) Replacing values with NaN
    (d) Converting to string

31. In supervised machine learning, what is the primary purpose of using labeled data during the training phase?
    (a) To introduce randomness in the model
    (b) To allow the model to learn the relationship between inputs and outputs
    (c) To increase the complexity of the model
    (d) To reduce the amount of data required for training

32. Which of the following algorithms is primarily used for classification tasks in supervised learning?

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(a) Linear Regression

(b) K-Nearest Neighbors

(c) Gradient Descent

(d) Principal Component Analysis

33. What is the main challenge associated with overfitting in supervised learning models?

(a) The model performs well on unseen data

(b) The model captures noise in the training data

(c) The model requires less labeled data

(d) The model has a high bias

34. In the context of supervised learning, what is the role of hyperparameter tuning?

(a) To randomly select features for the model

(b) To adjust the model's parameters that are not learned during training

(c) To eliminate the need for labeled data

(d) To reduce the size of the training dataset

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35. Which of the following metrics is NOT commonly used to evaluate the performance of a supervised learning model?

(a) Accuracy

(b) Recall

(c) Mean Squared Error

(d) Entropy

36. What is the primary difference between classification and regression tasks in supervised learning?

(a) Classification predicts categorical outcomes, while regression predicts continuous outcomes

(b) Classification requires more data than regression

(c) Regression is always more complex than classification

(d) Classification can only be performed on numeric data

37. In supervised learning, what is the purpose of splitting the dataset into training and test sets?

(a) To increase the overall size of the dataset

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(b) To evaluate the model's performance on unseen data

(c) To ensure all data points are used in training

(d) To reduce the complexity of the model

38. Which supervised learning algorithm is particularly known for its ability to handle both classification and regression tasks?

(a) Decision Trees

(b) K-Nearest Neighbors

(c) Naive Bayes

(d) Linear Regression

39. What is the main advantage of using ensemble methods like Random Forests in supervised learning?

(a) They simplify the model-building process

(b) They reduce the likelihood of overfitting by combining multiple models

(c) They require less computational power

(d) They eliminate the need for labeled data

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40. Which of the following statements correctly describes the concept of bias in supervised learning models?

(a) Bias refers to the model's inability to learn from the training data

(b) Bias is the error introduced by approximating a real-world problem with a simplified model

(c) Bias only affects the performance of regression models

(d) Bias is always desirable in model training

41. What is the primary objective of linear regression?

(a) To classify data into distinct categories

(b) To model the relationship between a dependent variable and one or more independent variables

(c) To reduce the dimensionality of data

(d) To cluster data points based on similarity

42. In simple linear regression which of the following represents the equation of the regression line?

(a) $y = mx + c$

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(b) $y = ax^2 + bx + c$

(c) $y = e^{(mx+c)}$

(d) $y = log(mx + c)$

43. What does the slope 'm' in the linear regression equation $y = mx + c$ indicate?

    (a) The point where the line crosses the y-axis

    (b) The rate of change of the dependent variable with respect to the independent variable

    (c) The maximum value of the dependent variable

    (d) The minimum value of the independent variable

44. In the context of linear regression what is the cost function commonly used to measure?

    (a) The accuracy of the model

    (b) The error between predicted and actual values

    (c) The complexity of the model

    (d) The number of features in the model

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45. Which optimization algorithm is commonly used to minimize the cost function in linear regression?

    (a) Gradient Descent

    (b) K-Means Clustering

    (c) Apriori Algorithm

    (d) Decision Trees

46. What is the key assumption about the relationship between the dependent and independent variables in linear regression?

    (a) Non-linear relationship

    (b) Exponential relationship

    (c) Logarithmic relationship

    (d) Linear relationship

47. What type of problems is Logistic Regression used for?

    (a) Regression Problems

    (b) Classification Problems

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(c) Dimensionality Reduction

(d) Clustering Problems

48. Which activation function is used in Logistic Regression?

(a) ReLU

(b) Sigmoid

(c) Softmax

(d) Tanh

49. What is the output of Logistic Regression?

(a) Continuous values

(b) Probability values between 0 and 1

(c) Categorical labels directly

(d) Clusters

50. Which type of Logistic Regression is used for ordered categories?

(a) Binomial

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(b) Multinomial

(c) Ordinal

(d) Linear

51. What is the primary difference between Random Forest Classifier and Random Forest Regressor?

(a) Classifier is used for numerical predictions, while Regressor is used for categorical predictions

(b) Classifier is used for categorical predictions, while Regressor is used for numerical predictions

(c) Both perform the same function

(d) Classifier is slower than Regressor

52. Which library is used to implement Random Forest in Python as shown in the articles?

(a) TensorFlow

(b) NumPy

(c) Scikit-learn

(d) Pandas

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